Aryan Roy

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Education

Georgia Institute of Technology — Atlanta, GA
Master of Science in Computer Science | *Machine Learning Specialization* | ***GPA: 4.0/4.0*** — *Exp: May 2027*

Manipal Institute of Technology — India
Bachelor of Technology | *Computer and Communication Engineering* | ***GPA: 8.52/10*** — *July 2023*

Publications

[ACL '26] **Grounded Concreteness: Human-Like Concreteness Sensitivity in Vision–Language Models**
A. Roy, Z. Wang, C. J. MacLellan. *Findings of the Association for Computational Linguistics: ACL 2026*

[CHI '26] **When Should Users Check? Modeling Confirmation Frequency in Multi-Step Agentic AI Tasks**
J. Zhou, **A. Roy**, S. Gupta, D. Weitekamp, C. J. MacLellan. *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2026*

[JPCS '26] **Using a DSL to Read ROOT TTrees Faster in Uproot**
A. Roy, J. Pivarski. *Journal of Physics: Conference Series, 3206(1), 2026*

[EPJ '25] **Parsing TTree Formula in Python**
A. Roy, J. Pivarski. *EPJ Web of Conferences, 337, 2025*

[JPCS '23] **An Array-Oriented Python Interface for FastJet**
A. Roy, J. Pivarski, C. W. Freer. *Journal of Physics: Conference Series, 2438(1), 2023*

Research & Work Experience

Vertically Integrated Projects, Georgia Tech — Atlanta, GA
Graduate Research Assistant — *August 2026 – Present*

- Researching methods for problem recommendation in introductory programming courses under **Dr. Rodrigo Valente**.
- Developing Knowledge Tracing and Mastery prediction models for optimal learning outcomes.
- Investigating methods for faithful student simulation using LLMs and developing benchmarks / metrics for quantifying progress.

University of Illinois Urbana-Champaign — Remote
Research Collaborator — *June 2026 – Present*

- Developing particle-based diffusion inference methods for guided generation with **Dr. Tong Zhang**, focusing on improving sample quality and adherence to guidance signals using rejection sampling and variance control methods.

MathWorks
EDG Intern — *May 2026 – August 2026*

- Built a multi-service AI analysis assistant with FastAPI, orchestrating LLM tool calls against a live, high-volume data service to produce aggregate insights at query time.
- Designed custom LLM tools and API clients for guided analysis, with eval harnesses to track answer quality and grounding across releases.

Teachable AI Lab, Georgia Tech — Atlanta, GA
Graduate Researcher — *July 2025 – Present*

- Developing a novel Vision model architecture for encoding and generating image data using sample-efficient hierarchical representation under **Dr. Christopher MacLellan**.
- Researching representation learning and interpretability in foundation models; designing controlled experiments on probing, activation steering, and concept attribution. Published at CHI and ACL.
- Fine-tuning and serving LLMs up to 90B parameters on Slurm clusters (4–8 A40 GPUs) using LoRA / PEFT, mixed-precision, and distributed training.

Oracle — India
Software Engineer — *January 2023 – August 2025*

- Delivered "Ask Oracle," a multi-data-source RAG system for business QnA; built retrieval, chunking, and re-ranking pipelines and iterated on quality via offline evals and human feedback.
- Built a production monitoring service for OCI Streaming + Kafka with Prometheus, surfacing latency, throughput, and reliability metrics for downstream consumers.
- Implemented an API activity monitoring service over the STS API ($>$1M hits/day), exposing usage distributions and reliability signals.

Fermi National Accelerator Lab Batavia, IL

Research Intern, Computing Division *May 2022 – July 2022*

- Integrated a lightweight experimental DSL into a widely-used Python I/O library for runtime parsing of serialized objects under **Dr. Lindsey Gray**; achieved 400x speedup on target data types.

Princeton University Remote

Research Intern *April 2021 – August 2021*

- Built a Python library for sequential-recombination clustering on particle-physics data under **Dr. Jim Pivarski**; 25x throughput vs. existing implementations.

Open Source

pywhy-graphs

Open-Source Contributor *January 2023 – May 2024*

- Implemented core algorithms for statistical causality (graph identification, SEMs) under **Dr. Adam Li**.

Uproot

Open-Source Contributor *January 2023 – December 2023*

- Designed and implemented a Python transpiler using Lark to translate between 3 standard scientific data formats in High Energy Physics.

Technical Skills

Languages: Python, SQL, Java, Bash

ML / Modeling: LLMs, VLMs, Representation Learning, RAG, Fine-tuning (LoRA / PEFT), Distributed Training, Evals & Ablations, Causal Inference, Statistical Modeling

Frameworks: PyTorch, Transformers (Hugging Face), scikit-learn, NumPy, pandas, FastAPI

Data & Infra: PostgreSQL, Kafka, Prometheus, Slurm, GitHub Actions, Conda, CI/CD

Awards & Activities

- **Graduate Research Assistant**: Competitive Research fellowships with full tuition waiver.
- **Graduate Teaching Assistant**: Multiple competitive teaching fellowships with full tuition waiver (CS 4635 A/7637).
- **IRIS-HEP Fellowship (2021, 2022)**: Awarded twice for HPC workflow contributions in High Energy Physics.
- **MITACS GRI (2022)**: Fully funded scholarship for neurologically-inspired authentication research (Declined).